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Assignment on Amortization

Amortization Schedule

A table showing the repayment schedule of interest and principal necessary to pay off a loan by maturity.

The calculation for annual loan installments (if loan installment is paid at the end of the year)

$$\text{Equated annual installment} = \text{loan amount} / \text{PVIFA}$$

PVIFA
Present value
interest factor

The calculation for annual loan installments (if loan installment is paid at the beginning of the year, $t = 0$)

$$\text{Annual installment of loan} = \text{loan amount} / \text{PVIFA} + 1.0$$

Example 1

Suppose you have made an arrangement of borrowing \$40. The interest rate is 20% on the borrowed amount. If we have to pay back the loan in five years, the equated annual installment will be calculated as:

$$\begin{aligned} \text{Equated annual installment} &= \text{loan amount} / \text{PVIFA (at 20\% for 5 years)} \\ &= \$40 / 2.990 = \$13.377 \end{aligned}$$

Example 2

Suppose you have made an arrangement of borrowing \$40. The interest rate is 20% on the borrowed amount. If we have to pay back the loan in five years, and the installment is due at the beginning of the year the equated annual installment will be calculated as:

$$\begin{aligned} \text{Equated annual installment} &= \text{loan amount} / (\text{PVIFA} + 1) \text{ (at 20\% for 4 years)} \\ &= \$40 / 2.588 + 1 = \$11.1 \end{aligned}$$

Assignment Questions

Activity 1: Set up an amortization schedule for a \$35,000 loan to be repaid in equal installments at the end of each of the next 5 years. The interest rate is 13%.

loan installment at the end of year
 $PVIFA = 3.5172$ $\text{loan amount} / PVIFA = 9951.097$

Year	Loan Installment	Beginning Balance	Interest Amount @ 13%	Principal Amount	Outstanding Balance
0					
1	9951.097	35000	$35000 \times 13 / 100 = 4550$	$9951.097 - 4550 = 5401.097$	$35000 - 5401.097 = 29598.903$
2	9951.097	29598.903	3847.74	$9951.097 - 3847.74 = 6103.357$	23495.54
3	9951.097	23495.54	3054.42	6896.67	16598.87
4	9951.097	16598.87	2157.85	7793.24	8805.63
5	9951.097	8805.63	1145.73	8805.63	0

$9951.097 - 8805.63$

Activity 2: Set up an amortization schedule for a \$75,000 loan to be repaid in equal installments at the end of each of the next 6 years. The interest rate is 12%.

$\text{loan amount} / PVIFA = 4.1114$
 $75000 / 4.1114 = 18241.96$

Year	Loan Installment	Beginning Balance	Interest Amount @ 12%	Principal Amount	Outstanding Balance
0					
1	18241.96	75000	9000	9241.96	65758.04
2	18241.96	65758.04	7891	10350.96	55407.08
3	18241.96	55407.08	6649	11592.96	43814.12
4	18241.96	43814.12	5258	12983.96	30830.16
5	18241.96	30830.16	3700	14541.96	16288.20
6	18241.96	16288.20	1955	16288.20	0

Activity 3: Set up an amortization schedule for a \$155,000 loan to be repaid in equal installments at the end of each of the next 8 years. The interest rate is 15%.

Loan amount / PVIFA

$$155,000 / 4.4873 = 34541.92$$

Year	Loan Installment	Beginning Balance	Interest Amount @ 15%	Principal Amount	Outstanding Balance
0					
1	34541.92	155000	23250	11291	143708.1
2	34541.92	143708.1	21556.21	12985.71	130722.38
3	34541.92	130722.3	19608.35	14933.57	115788.81
4	34541.92	115788.81	17368.32	17173.60	98615.20
5	34541.92	98615.20	14792.28	19749.62	78865.58
6	34541.92	78865.56	11829.83	22712.09	56153.46
7	34541.92	56153.46	8423.02	26130.44	30023.02
8	34541.92	30023.02	4503.45	29999.57	0

Activity 4: A \$250,000 loan at 25% annually is taken by a company in 2021. It is to be repaid in equal installments at the beginning of each year of the next 9 years. Prepare an amortization schedule for the payments.

$$\text{Loan amount} / (PVIFA + 1) = 250000 / (3.3289 + 1) = 57751.39$$

Year	Loan Installment	Beginning Balance	Interest Amount @ 25%	Principal Amount	Outstanding Balance
0	57751	250000	0	57751	250000 - 57751 = 192249
1	57751	192249	48062	9689	182560
2	57751	182560	45638	12112	170448
3	57751	170448	42612	15139	155309
4	57751	155309	38827	18924	136385
5	57751	136385	34096	23655	112730
6	57751	112730	28182	29569	83161
7	57751	83161	20790	36971	46190
8	57751	46190	11547	46190	0
9	57751				

Activity 5: A \$500,000 loan at 15% annually is taken by a company in 2018. It is to be repaid in equal installments at the beginning of each year of the next 4 years. Prepare an amortization schedule for the payments.

$$\text{Loan installment} = \frac{\text{Loan amount}}{(PVIFA + 1)}$$

$$\text{at beginning of year} = \frac{500,000}{(2.2832 + 1)} = 152290.44$$

Year	Loan Installment	Beginning Balance	Interest Amount @ 15%	Principal Amount	Outstanding Balance
0	152290	500,000	0	152290	500,000 - 152290 = 347710
1	152290	347710	52156	100134	247576
2	152290	247576	37136	115154	132422
3	152290	132422	19868	132422	0
4					
5					

23i-5013

BSBA-4B

Assignment on Amortization

Amortization Schedule

A table showing the repayment schedule of interest and principal necessary to pay off a loan by maturity.

The calculation for annual loan installments (if loan installment is paid at the end of the year)

$$\text{Equated annual installment} = \text{loan amount} / \text{PVIFA}$$

The calculation for annual loan installments (if loan installment is paid at the beginning of the year, $t = 0$)

$$\text{Annual installment of loan} = \text{loan amount} / \text{PVIFA} + 1.0$$

Example 1

Suppose you have made an arrangement of borrowing \$40. The interest rate is 20% on the borrowed amount. If we have to pay back the loan in five years, the equated annual installment will be calculated as:

$$\begin{aligned} \text{Equated annual installment} &= \text{loan amount} / \text{PVIFA} \text{ (at 20\% for 5 years)} \\ &= \$40 / 2.990 = \$13.377 \end{aligned}$$

from table
years 5
rate 20%
[*]

Example 2

Suppose you have made an arrangement of borrowing \$40. The interest rate is 20% on the borrowed amount. If we have to pay back the loan in five years, and the instalment is due at the beginning of the year the equated annual installment will be calculated as:

$$\begin{aligned} \text{Equated annual installment} &= \text{loan amount} / (\text{PVIFA} + 1) \text{ (at 20\% for 4 years)} \\ &= \$40 / 2.588 + 1 = \$11.1 \end{aligned}$$

in beginning of year
we minus 1 from
total years.

$$5 - 1 = 4$$

Assignment Questions

Activity 1: Set up an amortization schedule for a \$35,000 loan to be repaid in equal installments at the end of each of the next 5 years. The interest rate is 13%.

$$\text{loan installment at end of year} = \frac{\text{loan amount}}{\text{PVIFA}} = \frac{35000}{3.5172} = 9951.097$$

1st Jan ← 0
end of year 1 ← 1

Year	Loan Installment	Beginning Balance	Interest Amount @ 13%	Principal Amount	Outstanding Balance
0	-	35000	0	0	35000
1	9951.097	35000	$35000 \times 13/100 = 4550$	$9951.097 - 4550 = 5401.097$	$35000 - 5401.097 = 29598.903$
2	9951.097	29598.903	3847.857	6103.24	23495.663
3	9951.097	23495.663	3054.436	6896.661	16599.002
4	9951.097	16599.002	2157.870	7793.227	8805.775
5	9951.097	8805.775	$9951.097 - 8805.775 = 1145.322$	8805.775	0

Activity 2: Set up an amortization schedule for a \$75,000 loan to be repaid in equal installments at the end of each of the next 6 years. The interest rate is 12%.

$$\text{loan installment at the end of year} = \frac{\text{loan amount}}{\text{PVIFA}} = \frac{75000}{4.1114} = 18241.961 \approx 18241$$

Year	Loan Installment	Beginning Balance	Interest Amount @ 12%	Principal Amount	Outstanding Balance
0	-	75000	0	0	75000
1	18241	75000	9000	9241	65759
2	18241	65759	7891.08	10349.92	55409.08
3	18241	55409.08	6649.089	11591.911	43817.169
4	18241	43817.169	5258.060	12982.939	30834.229
5	18241	30834.229	3700.107	14540.892	16293.336
6	18241	16293.336	1947.664	16293.336	0

Activity 3: Set up an amortization schedule for a \$155,000 loan to be repaid in equal installments at the end of each of the next 8 years. The interest rate is 15%.

$$\text{Loan installment at the end of year} = \frac{\text{Loan amount}}{PVIFA} = \frac{155000}{4.4873} = 34541.929 \approx 34541$$

Year	Loan Installment	Beginning Balance	Interest Amount @ 15%	Principal Amount	Outstanding Balance
0		155000	0	0	155000
1	34541.929	155000	23250	11291.929	143708.071
2	34541.929	143708.071	21556.210	12985.719	130722.352
3	34541.929	130722.352	19608.352	14933.577	115788.775
4	34541.929	115788.775	17368.316	17173.613	98615.162
5	34541.929	98615.162	14792.274	19749.655	78865.507
6	34541.929	78865.507	11879.826	22712.103	56153.404
7	34541.929	56153.404	8423.00	26118.919	30034.485
8	34541.929	30034.485	4507.444	30034.485	0

Activity 4: A \$250,000 loan at 25% annually is taken by a company in 2021. It is to be repaid in equal installments at the beginning of each year of the next 9 years. Prepare an amortization schedule for the payments.

$$\text{Loan installment at beginning of year} = \frac{\text{Loan amount}}{PVIFA + 1.0} = \frac{250000}{3.3289 + 1.0} = 57751.391 \approx 57751$$

Year	Loan Installment	Beginning Balance	Interest Amount @ 25%	Principal Amount	Outstanding Balance
0	57751	250000	0	57751	250000 - 57751 = 192249
1	57751	192249	48062.25	9688.75	182560.25
2	57751	182560.25	45640.062	12110.938	170449.312
3	57751	170449.312	42612.328	15138.672	155310.64
4	57751	155310.64	38827.66	18923.34	136387.3
5	57751	136387.3	34096.825	23654.175	112733.125
6	57751	112733.125	28183.281	29567.719	83165.406
7	57751	83165.406	20791.351	36959.649	46205.757
8	57751	46205.757	11545.243	46205.757	0
9					

Activity 5: A \$500,000 loan at 15% annually is taken by a company in 2018. It is to be repaid in equal installments at the beginning of each year of the next 4 years. Prepare an amortization schedule for the payments.

$$\text{loan installments at beginning of year} = \frac{\text{loan amount}}{PVIFA + 1.0} = \frac{500,000}{2.2832 + 1.0} = 152,290.4483$$

Year	Loan Installment	Beginning Balance	Interest Amount @ 15%	Principal Amount	Outstanding Balance
0	152,290.448	500,000	0	152,290.448	500,000 - 152,290.448 = 347,709.552
1	152,290.448	347,709.552	52,156.432	100,134.016	247,575.536
2	152,290.448	247,575.536	37,136.330	115,154.118	132,431.418
3	152,290.448	132,431.418	19,869.03	132,431.418	0
4					
5					

23i-5107
Sultan Buland
BSBA - C

Assignment on Amortization

Amortization Schedule

A table showing the repayment schedule of interest and principal necessary to pay off a loan by maturity.

The calculation for annual loan installments (if loan installment is paid at the end of the year)

$$\text{Equated annual installment} = \text{loan amount} / \text{PVIFA}$$

The calculation for annual loan installments (if loan installment is paid at the beginning of the year, $t = 0$)

$$\text{Annual installment of loan} = \text{loan amount} / \text{PVIFA} + 1.0$$

Example 1

Suppose you have made an arrangement of borrowing \$40. The interest rate is 20% on the borrowed amount. If we have to pay back the loan in five years, the equated annual installment will be calculated as:

$$\begin{aligned} \text{Equated annual installment} &= \text{loan amount} / \text{PVIFA (at 20\% for 5 years)} \\ &= \$40 / 2.990 = \$13.377 \end{aligned}$$

Example 2

Suppose you have made an arrangement of borrowing \$40. The interest rate is 20% on the borrowed amount. If we have to pay back the loan in five years, and the instalment is due at the beginning of the year the equated annual installment will be calculated as:

$$\begin{aligned} \text{Equated annual installment} &= \text{loan amount} / (\text{PVIFA} + 1) \text{ (at 20\% for 4 years)} \\ &= \$40 / 2.588 + 1 = \$11.1 \end{aligned}$$

Assignment Questions

Activity 1: Set up an amortization schedule for a \$35,000 loan to be repaid in equal installments at the end of each of the next 5 years. The interest rate is 13%.

$$EAI = \text{Loan amount} / PVIFA \text{ (at 13\% of 5 years)}$$

$$= 35,000 / 3.5772$$

Year	Loan Installment	Beginning Balance	Interest Amount @ 13%	Principal Amount	Outstanding Balance
0					
1	9951.097	35,000	$35,000 \times 0.13 = 4,550$	5401.097	29,598.903
2	9951.097	29,598.903	3847.8739	6103.2394	23,495.663
3	9951.097	23,495.663	3054.436	6896.661	16,599.002
4	9951.097	16,599.002	2157.870	7793.227	8805.775
5	9951.097	8805.775	1145.322	8805.772	0

Activity 2: Set up an amortization schedule for a \$75,000 loan to be repaid in equal installments at the end of each of the next 6 years. The interest rate is 12%.

$$EAI = \text{Loan amount} / PVIFA \text{ (at 12\% for 6 years)}$$

$$= 75,000 / 4.114$$

Year	Loan Installment	Beginning Balance	Interest Amount @ 12%	Principal Amount	Outstanding Balance
0					
1	18241	75,000	9000	9241	65759
2	18241	65759	7891	10350	55409
3	18241	55409	6649	11592	43817
4	18241	43817	5258	12983	30834
5	18281	30834	3700	14541	16293
6	18281	16293	1948	16293	0

Activity 3: Set up an amortization schedule for a \$155,000 loan to be repaid in equal installments at the end of each of the next 8 years. The interest rate is 15%.

$$EIA = \text{Loan Amount} / PVIFA (\text{at } 15\% \text{ of } 8 \text{ years})$$

$$= 155,000 / 4.4873$$

Year	Loan Installment	Beginning Balance	Interest Amount @ 15%	Principal Amount	Outstanding Balance
0					
1	34541.929	155,000	23250	11291929	143708.071
2	34541.929	143708.071	21556.210	12985.719	130722.352
3	34541.929	130722.352	19608.352	14933.576	115788.775
4	34541.929	115788.775	17368.316	17173.612	98615.162
5	34541.929	98615.162	14792.274	19749.654	78865.507
6	34541.929	78865.507	11859.826	22712.102	56153.404
7	34541.929	56153.404	8423.016	26118.918	30034.485
8	34541.929	30034.485	4507.443	30034.485	0

Activity 4: A \$250,000 loan at 25% annually is taken by a company in 2021. It is to be repaid in equal installments at the beginning of each year of the next 9 years. Prepare an amortization schedule for the payments.

$$A1L = \text{Loan amount} / PVIFA + 1.0 (\text{at } 25\% \text{ for } 9 \text{ years})$$

$$= 250,000 / 3.3289 + 1$$

Year	Loan Installment	Beginning Balance	Interest Amount @ 25%	Principal Amount	Outstanding Balance
0	55751	250,000	0	55751	192249
1	55751	192249	48062	9689	182560
2	55751	182560	45640	12111	170449
3	55751	170449	42612	15138	155310
4	55751	155310	38827	18923	136386
5	55751	136386	34096	23654	112731
6	55751	112731	28182	29568	83162
7	55751	83162	20790	34960	48201
8	55751	48201	11549	46201	0
9					

Activity 5: A \$500,000 loan at 15% annually is taken by a company in 2018. It is to be repaid in equal installments at the beginning of each year of the next 4 years. Prepare an amortization schedule for the payments.

$$A_{IL} = \text{Loan Amount} / \text{FVIFA} (15\% \text{ for } 4 \text{ years})$$

$$= 500,000 / 2.2832 + 1$$

Year	Loan Installment	Beginning Balance	Interest Amount @ 15%	Principal Amount	Outstanding Balance
0	152290	500,000	0	152290	347710
1	152290	347710	52156	100134	247576
2	152290	247576	37136	115154	132422
3	152290	132422	19863	132422	0
4					
5					